



## THREE-PHASE HYBRID INVERTER

SUN-20 / 25 / 30 / 40 / 50K-SG01HP3-EU



## DATASHEET

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## 2. TECHNICAL SPECIFICATIONS

| Model                                | SUN-20K-SG01HP3-EU                      | SUN-25K-SG01HP3-EU | SUN-30K-SG01HP3-EU | SUN-40K-SG01HP3-EU | SUN-50K-SG01HP3-EU |
|--------------------------------------|-----------------------------------------|--------------------|--------------------|--------------------|--------------------|
| Battery Input Data                   |                                         |                    |                    |                    |                    |
| Battery Type                         | Lead-acid or Lithium-ion                |                    |                    |                    |                    |
| Battery Voltage Range                | 200~600V                                |                    |                    |                    |                    |
| Max. Charging Current                | 37A                                     | 37A+37A            |                    |                    |                    |
| Max. Discharging Current             | 37A                                     | 37A+37A            |                    |                    |                    |
| Charging Curve                       | 3 Stages/Equalization                   |                    |                    |                    |                    |
| Charging Strategy for Li-Ion Battery | Self-adaption to BMS                    |                    |                    |                    |                    |
| PV String Input Data                 |                                         |                    |                    |                    |                    |
| Max. DC Input Power                  | 26000W                                  | 32500W             | 39000W             | 52000W             | 65000W             |
| PV Input Voltage                     | 1000V                                   |                    |                    |                    |                    |
| Start-up Voltage                     | 160V                                    |                    |                    |                    |                    |
| MPPT Range                           | 200~850V                                |                    |                    |                    |                    |
| Full Load DC Voltage Range           | 360~850V                                | 365~850V           | 435~850V           | 450~850V           |                    |
| Rated DC Input Voltage               | 500V                                    | 625V               | 500V               | 500V               | 625V               |
| PV Input Current                     | 36A+36A                                 |                    | 36A+36A+36A        | 36A+36A+36A+36A    |                    |
| Max. PV I <sub>sc</sub>              | 50A+50A                                 |                    | 50A+50A+50A        | 50A+50A+50A+50A    |                    |
| No. of MPPT Trackers                 | 2                                       |                    | 3                  | 4                  |                    |
| No. of Strings Per MPPT Tracker      | 2+2                                     |                    | 2+2+2              | 2+2+2+2            |                    |
| AC Output Data                       |                                         |                    |                    |                    |                    |
| Rated AC Output and UPS Power        | 20000W                                  | 25000W             | 30000W             | 40000W             | 50000W             |
| Max. AC Power                        | 22000W                                  | 27500W             | 33000W             | 44000W             | 55000W             |
| AC Output Rated Current              | 30.3A                                   | 38A                | 45.6A              | 60.8A              | 75.8A              |
| Max AC Output Current                | 45.4A                                   | 41.8A              | 50.1A              | 66.9A              | 83.3A              |
| Max Continuous AC Passthrough        | 100A                                    |                    |                    |                    |                    |
| Power Factor                         | 0.8 leading to 0.8 lagging              |                    |                    |                    |                    |
| Output Frequency and Voltage         | 50-60Hz; 3L/N/PE 220/380Vac; 230/400Vac |                    |                    |                    |                    |
| Grid Type                            | Three Phase                             |                    |                    |                    |                    |
| DC Injection Current (mA)            | <0.5%I <sub>n</sub>                     |                    |                    |                    |                    |
| Efficiency                           |                                         |                    |                    |                    |                    |
| Max. Efficiency                      | 97.6%                                   |                    |                    |                    |                    |
| MPPT Efficiency                      | 97.0%                                   |                    |                    |                    |                    |
| Euro Efficiency                      | 99.9%                                   |                    |                    |                    |                    |
| Protection                           |                                         |                    |                    |                    |                    |
| PV Input Lightning Protection        | Integrated                              |                    |                    |                    |                    |
| Anti-islanding Protection            | Integrated                              |                    |                    |                    |                    |

|                                             |                                                                                                        |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------|
| PV String Input Reverse Polarity Protection | Integrated                                                                                             |
| Insulation Resistor Detection               | Integrated                                                                                             |
| Residual Current Monitoring Unit            | Integrated                                                                                             |
| Output Over Current Protection              | Integrated                                                                                             |
| Output Shorted Protection                   | Integrated                                                                                             |
| Output Over Voltage Protection              | Integrated                                                                                             |
| Surge Protection                            | DC Type II / AC Type II                                                                                |
| <b>Certifications and Standards</b>         |                                                                                                        |
| Grid Regulation                             | EN50549, AS4777.2, VDE0126, IEC61727, VDEN4105, G99, NBT32004, CEI0-21, NRS097, NBR16149/16150, RD1699 |
| Safety EMC / Standard                       | IEC62109-1/-2, EN61000-6-1, EN61000-6-2, EN61000-6-3, EN61000-6-4                                      |
| <b>General Data</b>                         |                                                                                                        |
| Operating Temperature Range                 | -25~60°C, >45°C Derating                                                                               |
| Cooling                                     | Smart Cooling                                                                                          |
| Noise                                       | <45dB                                                                                                  |
| Communication with BMS                      | RS485; CAN                                                                                             |
| Weight                                      | 44.5kg                                                                                                 |
| Size                                        | 647.5W × 537H × 303.5D                                                                                 |
| Protection Degree                           | IP65                                                                                                   |
| Installation Type                           | Wall-Mounted                                                                                           |
| Warranty                                    | 5 years                                                                                                |

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Full training support, manuals and videos on [www.sunsynk.com](http://www.sunsynk.com)



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Want to become a distributor? Or want to know more?

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